

Amit Gohel

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Professional Summary

AI/ML Engineer with 2+ years shipping production Python-based AI systems on enterprise data. Built multi-tenant RAG chatbots (LangChain, LangGraph, Milvus, GPT-4o) handling 50,000+ queries/month at 95% MRR, multi-model LLM pipelines across OpenAI, Anthropic, and Gemini, and on-premise document processing systems at scale. Designs end-to-end systems spanning retrieval, inference, and runtime infrastructure layers. Microsoft Azure AI Engineer Associate certified.

Work Experience

AI Engineer, Vivansh Infotech *Ahmedabad, India* Aug 2025 – Present

- Upgraded the **permission-aware RAG chatbot** to a fully **agentic architecture using LangChain** for a multi-tenant Environment, Health, and Safety (EHS) platform serving **500+ concurrent users** and **1,000+ queries/day** – integrated with **Milvus**, **PostgreSQL**, and **10+ platform APIs**, enabling the chatbot to autonomously retrieve live data, trigger workflows, and resolve multi-turn queries end-to-end
- Enforced **250+ granular permissions per query** in real time via 2-stage retrieval: dedicated Milvus permission index cross-referenced against PostgreSQL, with GPT-4o handling access control, query rewriting, and conversation resolution across 10,000+ content chunks; **80% MRR**
- Built an **AI-powered form builder** that converts PDF forms to digital replicas in ~20 seconds (**200+ forms generated in month 1**), with bulk processing of 100+ PDFs in 5–10 minutes – replacing 1+ hour of manual work per form
- Designed intermediate Pydantic-validated JSON schema enabling reliable Large Language Model (LLM)-to-platform conversion; implemented multi-model pipeline (GPT-4o, Qwen3-Coder-480B, GPT-4o-mini) with each model selected via empirical benchmarking
- Created an **AI video course generation pipeline** reducing production time from ~1 week to ~10 minutes (99% reduction) by orchestrating RAG-based video selection (Milvus + Cohere), Gemini 2.5 Pro for visual-audio alignment, ElevenLabs narration, and serverless FFmpeg rendering on AWS Lambda

AI/ML Developer, Silver Touch Technologies Ltd *Ahmedabad, India* Jun 2024 – Jul 2025

- Rebuilt production RAG chatbot infrastructure serving **50,000+ queries/month** across 10+ chatbot instances – migrated from in-code Llama-2-7b to VLLM-served API and replaced FAISS with Milvus, improving MRR by **111% (45% to 95%)**
- Led vector database evaluation (Milvus vs. Qdrant vs. FAISS) on production-scale data; tested 6+ embedding models (BGE-M3, BGE-small, BGE-base, BGE-large, Stella, SPLADE) and built custom retrieval logic with BGE-M3 + reranker
- Reduced chatbot response latency by **50% (16s to 8s)** through query preprocessing with intent identification and retrieval pipeline optimization
- Built **on-premise Optical Character Recognition (OCR) and document processing pipeline** for sensitive government documents (200+ page PDFs) – combined DocTR + DocLink OCR outputs via Llama-3.1-Nemotron-70b for hybrid merging, achieving ~90% entity extraction accuracy on Aadhaar, PAN, and GST numbers

Research Intern, Indian Institute of Technology (IIT) Delhi *New Delhi, India* Jan 2024 – May 2024

- Built end-to-end **air writing recognition system**: ESP32 + MPU6050 wearable capturing 6-axis Inertial Measurement Unit (IMU) data at 60Hz, classifying 62 alphanumeric characters via Convolutional Neural Network (CNN)-Bidirectional Long Short-Term Memory (BiLSTM) achieving 75% user-dependent accuracy
- Collected 12,400 labeled samples from 20 participants; systematically evaluated **1,000+ model configurations** across 5 architecture families (Dense, CNN, LSTM, BiLSTM, hybrids) using TensorFlow/Keras

Education

Bachelor of Engineering – Computer Engineering Jun 2020 – Jun 2024

Vishwakarma Government Engineering College, Ahmedabad, India

Technical Skills

- **Programming Languages:** Python, C++ , SQL
- **LLM Providers and APIs:** OpenAI (GPT-4o, GPT-4o-mini), Anthropic Claude, Google Gemini 2.5 Pro, Qwen3-Coder (Together AI), Cohere, ElevenLabs
- **AI/ML Frameworks:** LangChain, LangGraph, PyTorch, TensorFlow/Keras, Hugging Face Transformers, VLLM
- **Techniques:** Retrieval-Augmented Generation (RAG), Agentic AI, Prompt Engineering, Natural Language Processing (NLP), CNN, LSTM, BiLSTM
- **Vector Databases and Retrieval:** Milvus, FAISS, Qdrant, OpenAI Embeddings, BGE-M3, Cohere Embed/Rerank, SPLADE
- **Infrastructure and DevOps:** FastAPI, VLLM, Amazon Web Services (AWS) including EC2, Lambda, S3, SAM; Docker, Git, PostgreSQL, Pydantic, Selenium, FFmpeg
- **Hardware/IoT:** Arduino, ESP32, NodeMCU, IMU Sensors, Raspberry Pi

Certifications and Achievements

- **Microsoft Azure AI Engineer Associate** – Certified in designing and implementing Azure AI solutions
- **Samsung Solve for Tomorrow** – **Top 10 among 70,000+ participants nationwide**; featured on CNN News18
- **Smart India Hackathon 2023** – Top 5 among 44,000+ teams nationwide (Hardware Edition, MIT Karnataka)
- **Competitive Programming:** 370+ problems solved in Python